

Program: Diploma in Microelectronics	
Course Code: 5491	Course Title: Digital System Design
Semester: 5	Credits: 4
Course Category: Program core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To design of various combinational digital circuits using logic gates
- To analyze and design synchronous Sequential circuits
- To analyze and design asynchronous Sequential circuits
- To understand the concept of Hazards and design of Hazard free Circuit

Course Prerequisites:

Topic	Course code	Course name	Semester
Combinational Circuits		Digital Electronics	3
Sequential Circuits and its types		Digital Electronics	3
Hazards		Digital Electronics	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Design Various types of Combinational Circuits	15	Understanding
CO2	Analyze and Design Synchronous Sequential Circuit	15	Understanding
CO3	Analyze and Design Asynchronous Sequential Circuit	15	Understanding

CO4	Understand the various types of Hazards and design Hazard free Circuit	13	Understanding
	Series Tests	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3				
CO2	3	3	3				
CO3	3	3	3				
CO4	3	3	2				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Design Various types of Combinational Circuits		
M1.01	Design of different combinational Circuits	5	Apply
M1.02	Design combinational Circuits using Multiplexers and decoders	4	Apply
M1.03	Explain Different types of Adders	2	Understanding
M1.04	Design of Boolean function using Multiplexers and decoders	4	Apply
Contents: Problem formulation and design of combinational circuits - Code-Converters, Half and Full Adders, Magnitude Comparator, Decoder, Encoder, Priority Encoder, Multiplexer and Demultiplexer, Binary Parallel Adder – Carry lookahead Adder, BCD Adder. Combination logic design using Multiplexer and Decoder Case study: Digital trans-receiver / 8 bit Arithmetic and logic unit			
CO2	Design and Analyze Clocked Synchronous Sequential Network		

M2.01	Analyze the given Synchronous Circuit	4	Apply
M2.02	Design of Clocked Synchronous Sequential Circuit	4	Apply
M2.03	Design Various types of counters	4	Apply
M2.04	Design Various types of Registers	3	Apply
	Series Test 1	1	

Contents :

Flip flops – SR, JK, T, D, Master/Slave FF, Triggering of FF,

Clocked Synchronous Sequential Network (CSSN) -Analysis and design of clocked sequential circuits – Moore/Mealy models, state minimization, state assignment.

Circuit implementation - Counters, Ripple Counters, Sequence Generator, Ring Counters, Shift registers, Universal Shift Register.

Model Development: Designing of rolling display/real time clock

CO3	Design and Analyze Asynchronous Sequential Network		
M3.01	Analyze the given Asynchronous Circuit	4	Apply
M3.02	Explain stable state and unstable state and flow table with example	4	Understanding
M3.03	Explain State Assignment	4	Understanding
M3.04	Explain races and cycles in ASC	3	Understanding

Contents:

Analysis of Asynchronous Sequential Circuits (ASC), Flow table reduction- State assignment problem and the transition table, Stable State, Unstable state, Races and cycles in ASC

CO4	Design Hazard free Circuit		
M4.01	Explain Data synchronizer	3	Understanding
M4.02	Explain Static and Dynamic Hazards	3	Understanding

M4.03	Explain Essential Hazards	3	Understanding
M4.04	Explain MOM and Practical Issues	4	Understanding
	Series Test 2	1	
Contents : Hazards: Static Hazards, Dynamic Hazards, Essential Hazards, Pulse mode sequential circuits, Design of Hazard free circuits, Data synchronizer Mixed operating mode asynchronous circuits, Practical issues- clock skew and jitter, Synchronous and asynchronous inputs – switch bouncing			

Text/Reference:

T/R	Book Title/Author
T1	Digital Principles and Design, Donald D. Givone, McGraw Hill Higher Education, 2022
T2	Modern Digital Electronics – R. P. Jain, 3rd Edition, 2007- Tata McGraw-Hill
R1	Digital Design- Morris Mano, PHI, 4th Edition, 2006
R2	Charles H. Roth, Jr, ‘Fundamentals of Logic Design’, Jaico Books, 4th Edition, 2002.
R3	William I. Fletcher, “An Engineering Approach to Digital Design”, Prentice-Hall of India, 1980.
R4	John. F. Wakerly, “Digital Design Principles and Practices”, Pearson Education, 4th Edition, 2007.

Online Resources:

Sl. No.	Website Link
1	https://onlinecourses.nptel.ac.in/noc21_ee39
2	https://onlinecourses.nptel.ac.in/noc25_ee24/preview
3	https://www.geeksforgeeks.org/principles-in-digital-system-design/